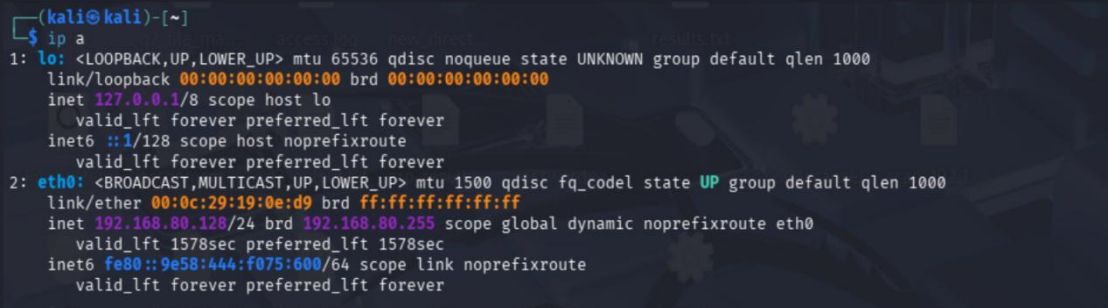


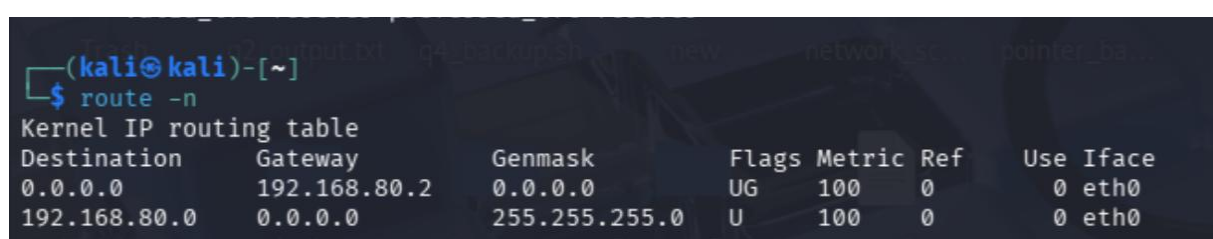
ASSIGNMENT

Section 3.2

1. 

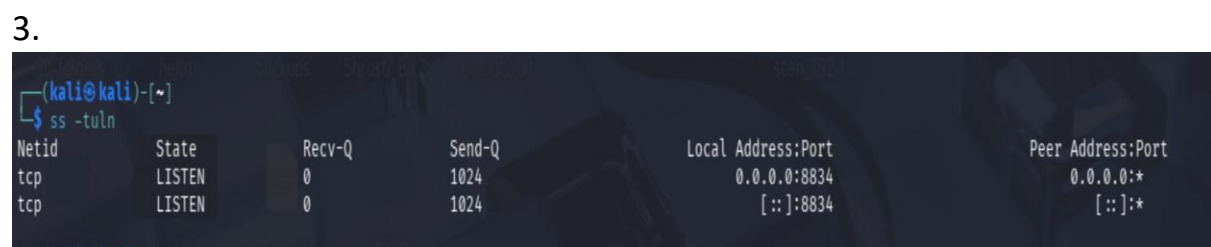
```
(kali@kali)~$ ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 00:0c:29:19:0e:d9 brd ff:ff:ff:ff:ff:ff
    inet 192.168.80.128/24 brd 192.168.80.255 scope global dynamic noprefixroute eth0
        valid_lft 1578sec preferred_lft 1578sec
    inet6 fe80::9e58:444:f075:600/64 scope link noprefixroute
        valid_lft forever preferred_lft forever
```

This command displays the link status and logical addresses for all network interfaces. The output identifies the loopback interface (lo) used for internal testing and the primary Ethernet interface (eth0), which is assigned the private IPv4 address 192.168.80.128.

2. 

```
(kali@kali)~$ route -n
Kernel IP routing table
Destination      Gateway          Genmask         Flags Metric Ref    Use Iface
0.0.0.0          192.168.80.2    0.0.0.0         UG    100    0      0 eth0
192.168.80.0     0.0.0.0         255.255.255.0   U     100    0      0 eth0
```

The routing table defines the path for data packets leaving the system to reach various destinations. The entry for 0.0.0.0 points to the gateway address 192.168.80.2, establishing the mandatory "next hop" for any traffic intended for the external internet.

3. 

```
(kali@kali)~$ ss -tln
Netid      State      Recv-Q      Send-Q      Local Address:Port      Peer Address:Port
tcp        LISTEN     0            1024        0.0.0.0:8834             0.0.0.0:*
tcp        LISTEN     0            1024        [::]:8834                [::]:*
```

This command lists active TCP and UDP connections currently in a LISTEN state. The results show that the system is monitoring port 8834 on both IPv4 and IPv6 stacks, indicating a local service is ready to accept incoming network requests.

- 4.
- ```
(kali㉿kali)-[~]
$ ping -c 4 google.com
PING google.com (142.250.67.46) 56(84) bytes of data.
64 bytes from pnmaaa-bb-in-f14.1e100.net (142.250.67.46): icmp_seq=1 ttl=128 time=39.8 ms
64 bytes from pnmaaa-bb-in-f14.1e100.net (142.250.67.46): icmp_seq=2 ttl=128 time=60.2 ms
64 bytes from pnmaaa-bb-in-f14.1e100.net (142.250.67.46): icmp_seq=4 ttl=128 time=56.3 ms

— google.com ping statistics —
4 packets transmitted, 3 received, 25% packet loss, time 3032ms
rtt min/avg/max/mdev = 39.772/52.085/60.195/8.851 ms
```

The ping utility verifies end-to-end connectivity by sending ICMP Echo Request packets to google.com. The results confirm successful name resolution to 142.250.67.46 and measure a mean latency of 52.085ms, though a 25% packet loss suggests minor network congestion or instability.

- 5.
- ```
(kali㉿kali)-[~]  
$ traceroute google.com  
traceroute to google.com (142.250.67.46), 30 hops max, 60 byte packets  
1 192.168.80.2 (192.168.80.2) 0.701 ms 0.418 ms 0.340 ms  
2 * * *  
3 * * *  
4 * * *  
5 * * *  
6 * * *  
7 * * *  
8 * * *  
9 * * *  
10 * * *  
11 * * *  
12 * * *  
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26 * * *  
27 * * *  
28 * * *  
29 * * *  
30 * * *
```

The traceroute identifies the sequential routers (hops) data takes to reach its destination. While the first hop successfully identifies the local gateway at 192.168.80.2, subsequent hops return asterisks (*), indicating that those routers are filtering the diagnostic packets for security purposes.

Parameter	Value
IP Address	192.168.80.128
Default Gateway	192.168.80.2
DNS Server	192.168.80.2